

E-Commerce performance Analysis

Overview

This project presents an in-depth analysis of e-commerce performance metrics, focusing on sales trends, profitability, customer demographics, and regional insights. The dashboards leverage data analytics and visualization tools to uncover valuable insights that drive business decisions.

Key Objectives:

- ✓ Analyze revenue, profit, and order trends over multiple years.
- ✓ Identify high-performing product categories and customer preferences.
- ✓ Assess sales distribution across regions and customer demographics.
- ✓ Provide actionable insights for business growth and optimization.

Tools & Technologies Used:

- Power BI & Tableau for interactive visualizations.
- ETL techniques for data cleaning and transformation.

Data Source: Kaggel

-Number of Columns: 20

-Number of Records: 113036

Key attributes :

Number of orders , Revenue , Profit, Country , Date , Customer Age , Customer Gender, Product Category , Product Name , product cost , Unit price

Cleaning & Transformation:

- No missing values were found in this dataset
- Remove duplicates and errors
- New Column was added to calculate the Profit Margin

- **Tools Used:** Power Bi

- **Data Modeling & DAX**

- **Data Model Structure:** The dataset includes only one table with 2 columns

- **DAX Measures & Calculations:**

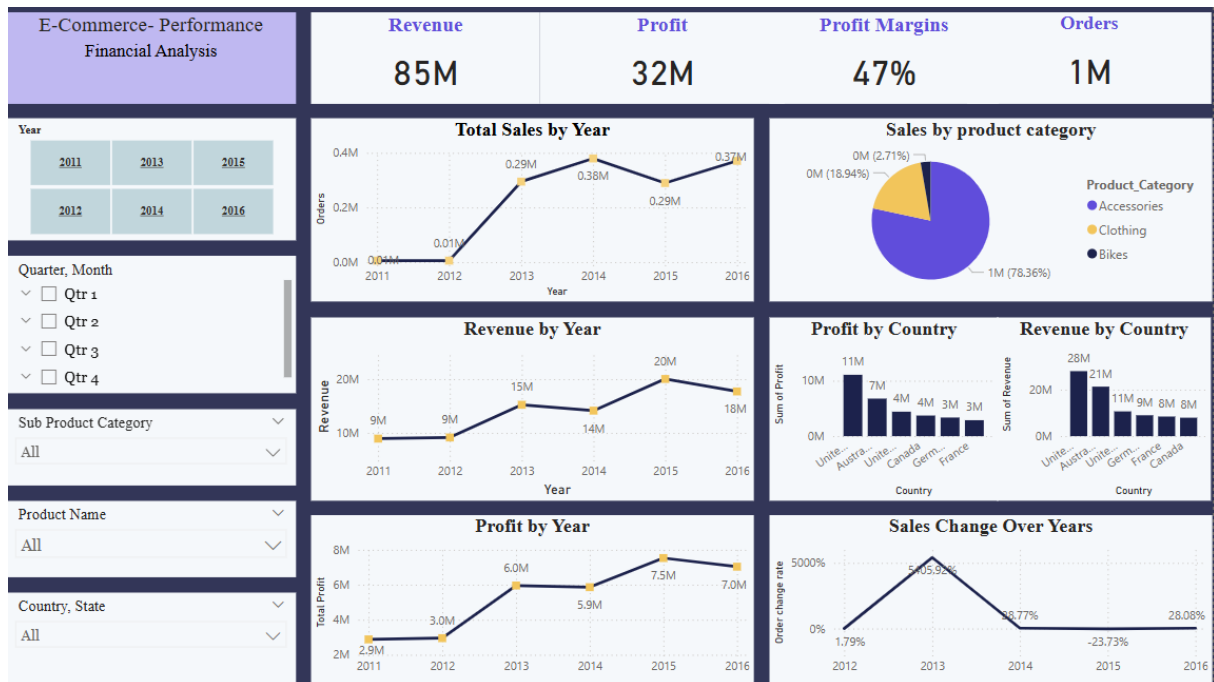
DAX was used to find the Maximum values for values, for example, most sold product :

- DAX was used to find the changes from 2011 to 2016.

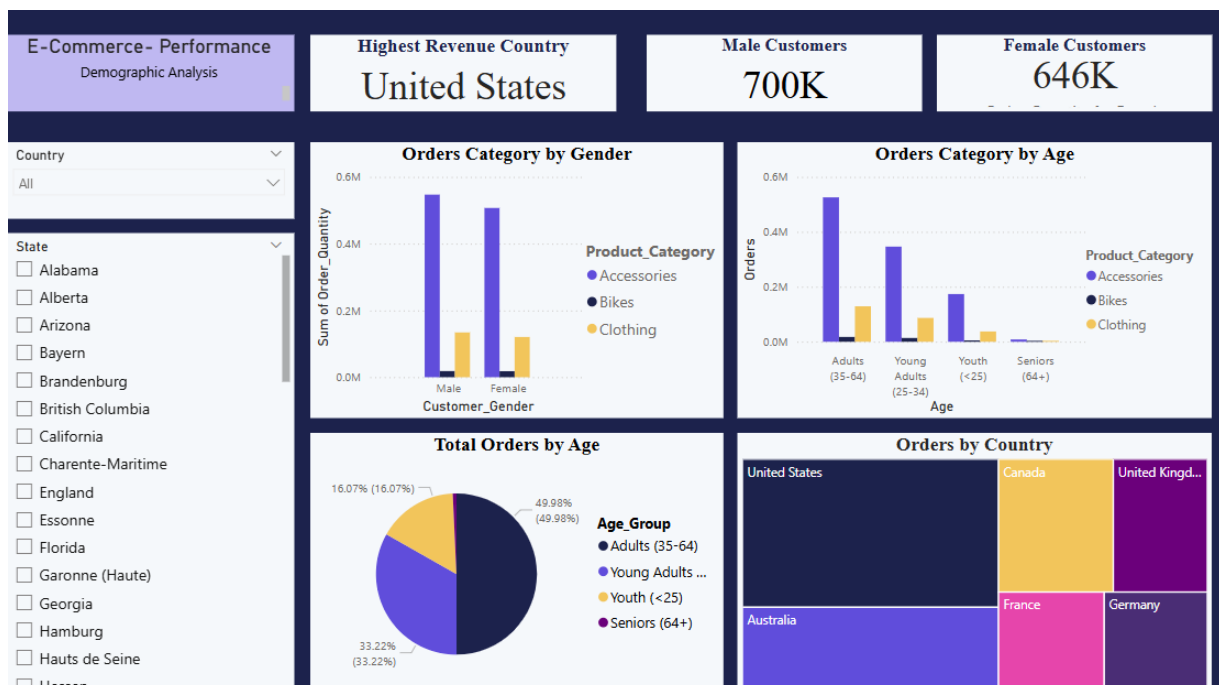
- Function was used to create a new column to calculate the Profit Margins.

4. Visualizations & Insights

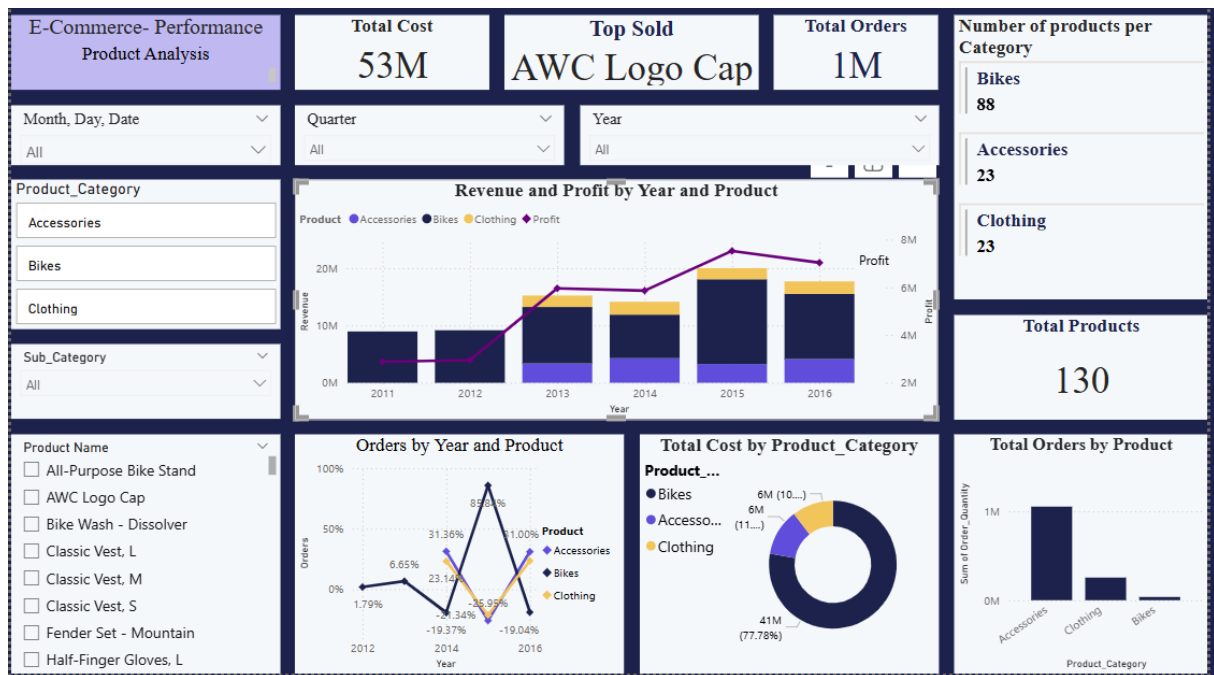
▪ Dashboard Overview:



Figur 1-Financial Performance



Figuur 2-Demographic Analysis



Figur 3-Product performance

Observations:

1. **Product Mix:** Bikes are your superstar, contributing to 78.36% of total sales. Accessories and Clothing play smaller roles.
2. **Geographic Distribution:** The United States and Australia are your top performers in both revenue and profit. This indicates a strong market presence in these countries.
3. **Revenue and Profit Trends:** Both metrics show a general upward trend, indicating growth. However, there was a slight decline in revenue and profit in 2016, which might need further investigation.
4. **Order Volatility:** The number of orders shows significant changes, especially in 2013, where there's a massive increase. Understanding the factors behind this surge could be crucial.

Business Recommendations

- 🔧 Lower bike production costs while maintaining quality.
- 📢 Prioritize marketing for the AWC Logo Cap and explore similar product opportunities.
- 📦 Expand the Accessories & Clothing categories to diversify profits.

-  Analyze the post-2015 sales decline and adjust strategies accordingly.
-  Promote high-demand, lower-cost Accessories to boost profit margins.
-  Implement targeted promotions for Clothing & Accessories to balance sales distribution.
-  Strengthen online marketing efforts for Young Adults (25-34) and Adults (35-64).
-  Enhance international presence in Canada & the U.K.
-  Utilize the gender-balanced audience for customized marketing campaigns.

Challenges:

As first non-academic project , there were some challenges including :

- Balancing aesthetic appeal with clarity and usability.
- Ensuring intuitive navigation and easy-to-understand insights.
- Choosing the right visualization types to represent data accurately.
- Even though public PowerBi offers many functions, some limitations were faced, including requesting a work school email address to publish and add more visualizations.

Solutions:

Overcoming these challenges required a mix of data preprocessing techniques, performance-tuning strategies, and user-centered design principles. in addition, coming up with project required alot of research and best practice

Conclusion

The E-Commerce Performance Financial Analysis dashboard paints a clear picture of your business's trajectory and potential. Overall, you've seen significant growth over the years, with both revenue and profit showing an upward trend.

- Bikes are the standout product, driving the majority of your sales.
- United States and Australia are your strongest markets, contributing significantly to your revenue and profit.
- The slight decline in 2016 and the order volatility warrant further investigation to ensure continued growth and stability.

With strategic focus on expanding in high-performing markets and leveraging the popularity of your best-selling product categories, your e-commerce business is poised for even greater success. Keep an eye on those few areas of concern, and you'll be well on your way to achieving your financial goals.

8. Links & Access

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